Marketing Campaign Budget Allocation under Operational Constraints

Motivation

Companies must allocate limited marketing budgets across multiple campaigns operating in different advertising channels. Each campaign differs in cost, expected return, risk level, and audience reach. Heuristic decisions may lead to inefficient spending or excessive risk exposure. This project proposes a mathematical optimization framework to systematically select the most profitable campaign portfolio under realistic business constraints.

Problem Statement and Decision Variables

Let $i = 1, \dots, n$denote available marketing campaigns. Each campaign is defined by:

- $Cost_i$: investment cost
- ROI_i: expected return coefficient
- $Risk_i$: risk score
- $Reach_i$: audience reach
- $Channel_i$: advertising channel

The company has a total marketing budget B.

Decision Variables

Let $i = 1, \dots, n$,denote the set of available marketing campaigns, each characterized by investment cost $Cost_i$, expected return coefficient ROI_i, risk score $Risk_i$, audience reach $Reach_i$, and channel classification. The binary decision variable x_i takes value 1 if campaign i is selected and 0 otherwise. The objective is to maximize total risk-adjusted expected return, formulated as $\max \sum_{i=1}^{n} ROI_i x_i - \lambda \sum_{i=1}^{n} Risk_i x_i$, where $\lambda \geq 0$ represents the decision maker's risk sensitivity level. The model is subject to operational constraints including a total budget limit $\sum_{i=1}^{n} Cost_i x_i \leq B$, channel capacity restrictions $\sum_{i \in C_k} x_i \leq L_k$for each channel k, and a minimum reach requirement $\sum_{i=1}^{n} Reach_i x_i \geq R_{min}$. The binary condition $x_i \in \{0,1\}$ensures discrete campaign selection decisions.

Which Method I Expect

The problem will be formulated as a MILP model due to binary decision variables and linear structure. The model will be implemented in Python using a MILP solver (Preferrably Gurobi or PuLP), and sensitivity analysis will be conducted over budget and risk parameters.

References (Paper Link or Domain Source)

- Bertsimas, D., & Tsitsiklis, J. (1997). *Introduction to Linear Optimization*. Athena Scientific.
- Martello, S., & Toth, P. (1990). *Knapsack Problems: Algorithms and Computer Implementations*. Wiley.
- Marketing budget allocation overview: https://www.investopedia.com/terms/m/marketing-budget.asp